

PASP User Manual

Version 0.4.3

1. Introduction

PASP is a lightweight, local statistical analysis tool designed for the terminal. It is optimized for mobile environments like Termux but works on any system with Python.

2. Installation

General:

```
pip install git+https://github.com/ppmena/pasp
```

3. Usage Styles

Style A: `pasp data.csv --auto`

Style B: `pasp anova data.csv var group`

4. Commands Reference

- **descriptives:**

Calculate mean, median, SD, and classify variables as Nominal, Ordinal, or Scale.

- **ttest-one:**

One-sample T-test with automatic Shapiro-Wilk normality fallback.

- **ttest-ind:**

Independent samples T-test with Levene homogeneity and comparative histograms.

- **ttest-paired:**

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Paired samples T-test with differences normality evaluation.

- **anova:**

One-way ANOVA with strict residuals normality, Levene check, and Bonferroni Post-Hoc.

- **correlation:**

Bivariate Pearson/Spearman correlation matrices.

- **regression:**

Simple linear OLS regression with preliminary descriptives and scatter plots.

- **chi-square:**

Chi-square test of independence for categorical variables with Cramer's V and standardized residuals.

- **doctor:**

Diagnostic report of dependencies and system environment.

- **update:**

Upgrade PASP dynamically to the latest version from GitHub.

5. Examples with Built-in Datasets

```
pasp examples/ejemplo_anova.csv --auto
```

```
pasp examples/ejemplo_ttest.csv --ttest-ind recovery_time treatment --plot
```

```
pasp examples/ejemplo_regresion.csv --regression exam_score hours_studied
```